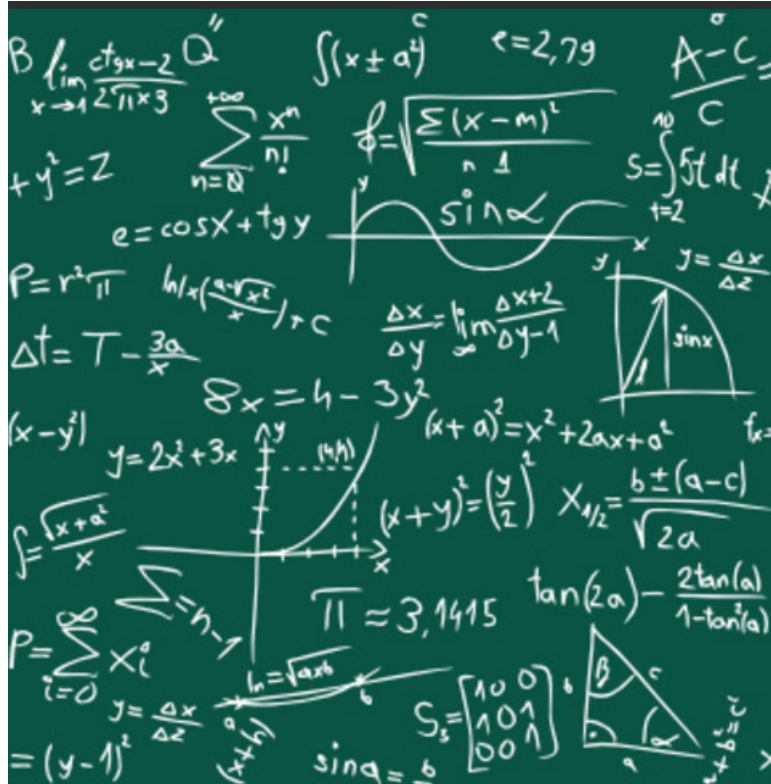


Computing derivative functions



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Table of the derivatives of the elementary functions:

	f	f'
1)	c (constant)	0
2)	x	1
3)	x^2	$2x$
4)	$\frac{1}{x}$	$-\frac{1}{x^2}$
5)	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$ (for $x > 0$)
6)	x^α ($\alpha \in \mathbb{R}, x > 0$)	$\alpha x^{\alpha-1}$
7)	$\sin x$	$\cos x$
8)	$\cos x$	$-\sin x$
9)	$\tan x$	$1 + \tan^2 x = \frac{1}{\cos^2 x}$
10)	$\cot x$	$-(1 + \cot^2 x) = -\frac{1}{\sin^2 x}$
11)	e^x	e^x
12)	$\log x$	$\frac{1}{x}$
13)	a^x ($a > 0$)	$a^x \log a$
14)	$\log_a x$ ($a > 0, a \neq 1$)	$\frac{1}{x \log a}$
15)	$\sinh x$	$\cosh x$
16)	$\cosh x$	$\sinh x$
17)	$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$
18)	$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$
19)	$\arctan x$	$\frac{1}{1+x^2}$

Exercise 1

Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3$.

1. Write the difference quotient of f at the point $x = 1$.
2. Compute $f'(1)$ as the limit of the difference quotient.
3. Write the equation of the tangent line to the graph of f at the point $(1, 1)$.

1.
$$\frac{f(1+h) - f(1)}{h} = \frac{(1+h)^3 - 1}{h} = \frac{3h + 3h^2 + h^3}{h} = 3 + 3h + h^2$$

2.
$$f'(1) = \lim_{h \rightarrow 0} 3 + 3h + h^2 = 3$$

3. The tangent line has equation $y = f(1) + f'(1)(x - 1)$, hence

$$y = 1 + 3(x - 1)$$

from which

$$y = 3x - 2$$

Exercise 2

Compute the derivative of the following function f , specifying the domain of f and of f' :

$$f(x) = x^2 \sin(x^3)$$

The functions f and f' are defined on all of $\mathbb{R}$.

Using the rules for the derivative of a product and of composite functions, we have

$$f'(x) = 2x \cdot \sin(x^3) + x^2 \cdot \cos(x^3) \cdot 3x^2 = 2x \sin(x^3) + 3x^4 \cos(x^3)$$

Exercise 3

Compute the derivative of the following function f , specifying the domain of f and of f' :

$$f(x) = x^3 \cos(e^{5x^2})$$

The functions f and f' are defined on all of $\mathbb{R}$.

Using the rules for the derivative of a product and of composite functions, we have

$$\begin{aligned} f'(x) &= 3x^2 \cdot \cos(e^{5x^2}) - x^3 \cdot \sin(e^{5x^2}) \cdot e^{5x^2} \cdot 10x = \\ &= 3x^2 \cos(e^{5x^2}) - 10x^4 e^{5x^2} \sin(e^{5x^2}) \end{aligned}$$

Exercise 4

Compute the derivative of the following function f , specifying the domain of f and of f' :

$$f(x) = e^{1/x}$$

Using the rules for the derivative of composite functions, we have

$$f'(x) = -\frac{1}{x^2} \cdot e^{1/x}$$

The functions f and f' are defined on $\mathbb{R} \setminus \{0\}$.

Exercise 5

Compute the derivative of the following function:

$$f(x) = 3x^4 + 5x + x^{3/2} - 2x^{-3}$$

$$f'(x) = 12x^3 + 5 + \frac{3}{2}\sqrt{x} + \frac{6}{x^4}$$

Exercise 6

Compute the derivatives of the following functions:

$$(1) f(x) = \log(|x|), \quad (2) f(x) = \log(3x), \quad (3) f(x) = \log\left(\left|\frac{x+2}{3-x}\right|\right)$$

1.

$$(\log(|x|))' = \frac{1}{|x|} \cdot \operatorname{sgn}(x) = \frac{1}{x}$$

2.

$$(\log(3x))' = (\log 3 + \log x)' = \frac{1}{x}$$

3.

$$\begin{aligned} \left(\log \left(\left| \frac{x+2}{3-x} \right| \right) \right)' &= \frac{1}{\left| \frac{x+2}{3-x} \right|} \cdot \operatorname{sgn} \left(\frac{x+2}{3-x} \right) \cdot \left(\left(\frac{x+2}{3-x} \right) \right)' = \frac{\left(\left(\frac{x+2}{3-x} \right) \right)'}{\frac{x+2}{3-x}} \\ &= \left(\left(\frac{x+2}{3-x} \right) \right)' \cdot \frac{3-x}{x+2} = \frac{3-x+x+2}{(3-x)^2} \cdot \frac{3-x}{x+2} \\ &= \frac{5}{(3-x) \cdot (x+2)} = \frac{5}{-x^2+x+6} \end{aligned}$$

Exercise 7

Compute the derivative of the following function:

$$f(x) = e^{-3x} (x^2 + 2x - 1)$$

Using the rules for the derivative of a product and of composite functions, we have:

$$\begin{aligned} f'(x) &= -3e^{-3x} (x^2 + 2x - 1) + e^{-3x} (2x + 2) = \\ &= e^{-3x} (-3x^2 - 6x + 3 + 2x + 2) = e^{-3x} (-3x^2 - 4x + 5) \end{aligned}$$

Exercise 8

Compute the derivative of the following function:

$$f(x) = \frac{1}{a} \arctan \left(\frac{x}{a} \right), \quad a > 0$$

$$f'(x) = \frac{1}{a} \cdot \frac{1}{1 + x^2/a^2} \cdot \frac{1}{a} = \frac{1}{a^2} \cdot \frac{1}{\frac{a^2+x^2}{a^2}} = \frac{1}{a^2 + x^2}$$

Exercise 9

Compute the derivative of the following function:

$$f(x) = x \log x$$

Using the rules for the derivative of a product, we have:

$$f'(x) = \log x + x \cdot \frac{1}{x} = \log x + 1.$$

Exercise 10

Compute the derivative of the following function:

$$f(x) = \arctan\left(\frac{1+x}{1-x}\right)$$

Using the rules for the derivative of a quotient and of composite functions, we have:

$$\begin{aligned} f'(x) &= \frac{1}{1 + \frac{(1+x)^2}{(1-x)^2}} \cdot \frac{(1-x) + (1+x)}{(1-x)^2} = \frac{(1-x)^2}{(1-x)^2 + (1+x)^2} \cdot \frac{2}{(1-x)^2} = \\ &= \frac{2}{1+x^2-2x+1+x^2+2x} = \frac{2}{2(x^2+1)} = \frac{1}{x^2+1} \end{aligned}$$

Exercise 11

Compute the derivative of the following function:

$$f(x) = e^{2x} (2 \sin 3x - 4 \cos 3x)$$

Using the rules for the derivative of a product and of composite functions, we have

$$\begin{aligned} f'(x) &= 2e^{2x} (2 \sin 3x - 4 \cos 3x) + e^{2x} (2 \cdot 3 \cos 3x + 4 \cdot 3 \sin 3x) = \\ &= 2e^{2x} (2 \sin 3x - 4 \cos 3x + 3 \cos 3x + 6 \sin 3x) = 2e^{2x} (8 \sin 3x - \cos 3x) \end{aligned}$$

Exercise 12

Compute the derivative of the following function:

$$f(x) = e^{\frac{x+2}{x-3}}$$

Using the rules for the derivative of composite functions, we have

$$f'(x) = e^{\frac{x+2}{x-3}} \left[\frac{(x-3) - (x+2)}{(x-3)^2} \right] = \frac{-5 e^{\frac{x+2}{x-3}}}{(x-3)^2}$$

Exercise 13

Compute the derivative of the following function:

$$f(x) = \tanh x$$

We compute the derivative of the hyperbolic tangent function

$$f(x) = \tanh x = \frac{\sinh x}{\cosh x}$$

Using the rules for the derivative of a quotient, we have

$$f'(x) = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = 1 - \tanh^2 x$$

Moreover, since $\cosh^2 x - \sinh^2 x = 1$, we also have

$$f'(x) = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x,$$

where $\operatorname{sech} x$ is the hyperbolic secant, defined as $\operatorname{sech} x = \frac{1}{\cosh x}$.

Exercise 14

Compute the derivative of the following function:

$$f(x) = \coth x$$

We compute the derivative of the hyperbolic cotangent function

$$f(x) = \coth x = \frac{\cosh x}{\sinh x}$$

Using the rules for the derivative of a quotient, we have

$$f'(x) = \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} = 1 - \coth^2 x$$

Moreover, since $\cosh^2 x - \sinh^2 x = 1$, we also have

$$f'(x) = -\frac{\cosh^2 x - \sinh^2 x}{\sinh^2 x} = -\frac{1}{\sinh^2 x} = -\operatorname{csch}^2 x,$$

where $\operatorname{csch} x$ is the hyperbolic cosecant, defined as $\operatorname{csch} x = \frac{1}{\sinh x}$.

Exercise 15

Compute the derivative of the following function:

$$f(x) = e^{x^2+3x}$$

Using the rules for the derivative of composite functions, we have

$$f'(x) = e^{x^2+3x} \cdot (2x+3)$$

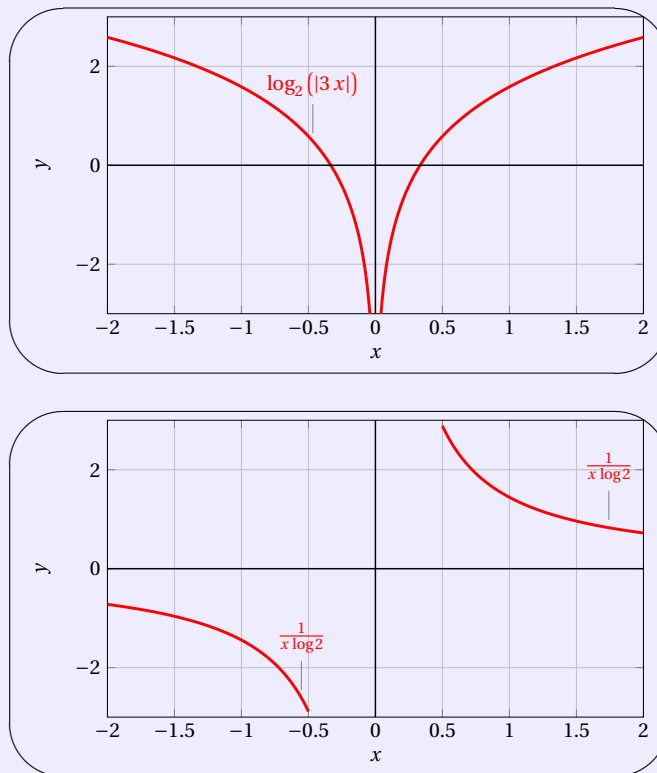
Exercise 16

Compute the derivative of the following function:

$$f(x) = \log_2(|3x|)$$

For $x \neq 0$, we have:

$$f(x) = \begin{cases} \log_2(3x), & x > 0 \\ \log_2(-3x), & x < 0 \end{cases} \quad \text{hence} \quad f'(x) = \begin{cases} \frac{3}{3x \log 2} = \frac{1}{x \log 2}, & x > 0 \\ \frac{-3}{-3x \log 2} = \frac{1}{x \log 2}, & x < 0 \end{cases}$$



The function is discontinuous at $x = 0$, hence it is not differentiable.

Exercise 17

Compute the derivative of the following function:

$$f(x) = \frac{x^2 + 3x - 2}{2x + 1}$$

Using the rules for the derivative of a quotient, we have

$$\begin{aligned} f'(x) &= \frac{(2x+3)(2x+1) - 2(x^2+3x-2)}{(2x+1)^2} = \frac{4x^2+2x+6x+3-2x^2-6x+4}{(2x+1)^2} \\ &= \frac{2x^2+2x+7}{(2x+1)^2} \end{aligned}$$

Exercise 18

Compute the derivative of the following function:

$$f(x) = x^2 \log(\cos x)$$

Using the rules for the derivative of composite functions, we have

$$f'(x) = 2x \cdot \log(\cos x) + x^2 \cdot \frac{1}{\cos x} \cdot (-\sin x) = 2x \log(\cos x) - x^2 \tan x$$

Exercise 19

Compute the derivative of the following function:

$$f(x) = \sqrt{\arctan(1 + x^2)}$$

Using the rules for the derivative of composite functions, we have

$$\begin{aligned} f'(x) &= \frac{1}{2\sqrt{\arctan(1 + x^2)}} \cdot \frac{1}{1 + (1 + x^2)^2} \cdot 2x = \frac{x}{(1 + (1 + x^2)^2)\sqrt{\arctan(1 + x^2)}} \\ &= \frac{x}{(x^4 + 2x^2 + 2)\sqrt{\arctan(1 + x^2)}} \end{aligned}$$

Exercise 20

Determine whether the function

$$f(x) = e^{-x+1} - 2\pi x - 1$$

is invertible and compute the derivative of $f^{-1}(y)$ for $y = -2\pi$.

Since $f'(x) = -e^{-x+1} - 2\pi < 0$, for every $x \in \mathbb{R}$, we immediately obtain that f is strictly monotonically decreasing and hence invertible on $\mathbb{R}$. We also observe that

$$e^{-x+1} - 2\pi x - 1 = -2\pi \iff x = 1,$$

i.e., $f^{-1}(-2\pi) = 1$. From the theorem on the derivative of the inverse function we immediately obtain

$$(f^{-1})'(-2\pi) = \frac{1}{f'(1)} = -\frac{1}{1+2\pi}.$$

Remark: We would not have been able to write the analytic expression of f^{-1} , even though it exists since f is invertible; the theorem on the derivative of the inverse function allows us to overcome this obstacle, without going through the explicit expression of the inverse.

Exercise 21

Determine whether the function

$$f(x) = 4x + \pi \sin x$$

is invertible and compute the derivative of $f^{-1}(y)$ for $y = 4\pi$.

Since $f'(x) = 4 + \pi \cos x > 0$, for every $x \in \mathbb{R}$, we immediately obtain that f is strictly monotonically increasing and hence invertible on $\mathbb{R}$. We also observe that

$$4x + \pi \sin x = 4\pi \iff x = \pi,$$

i.e., $f^{-1}(4\pi) = \pi$. From the theorem on the derivative of the inverse function we immediately obtain

$$(f^{-1})'(4\pi) = \frac{1}{f'(\pi)} = \frac{1}{4-\pi}.$$

Remark: We would not have been able to write the analytic expression of f^{-1} , even though it exists since f is invertible; the theorem on the derivative of the inverse function allows us to overcome this obstacle, without going through the explicit expression of the inverse.

Exercise 22

Let $f \in \mathcal{C}^1(\mathbb{R})$ be such that $f'(1) = 5e$. Setting $g(x) = f(\log x)$, compute $g'(e)$.

We observe that $g \in \mathcal{C}^1(0, +\infty)$, since it is a composition of functions of class $\mathcal{C}^1$ on their domains. Therefore, using the theorem on the derivative of composite functions, we obtain

$$g'(x) = f'(\log x) \frac{1}{x} \implies g'(e) = \frac{f'(1)}{e} = 5.$$

Exercise 23

Compute the derivative of the following function:

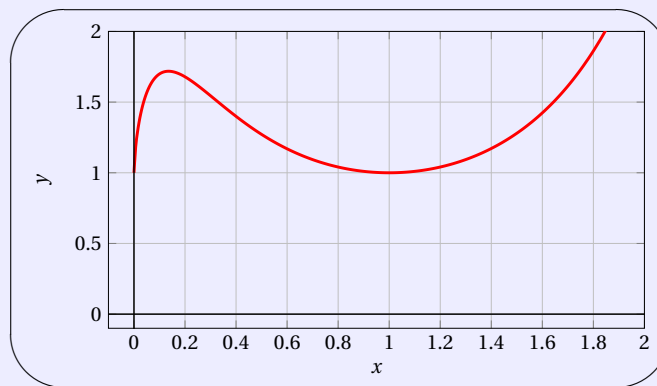
$$h(x) = x^{x \log x}$$

We use the rule for the derivative of a function raised to another function. The two functions are:

$$f(x) = x \quad \text{and} \quad g(x) = x \log x$$

Hence:

$$\begin{aligned} h'(x) &= \left(\exp(x \cdot \log^2 x) \right)' = \exp(x \cdot \log^2 x) \cdot (x \cdot \log^2 x)' = \\ &= x^{x \log x} \cdot \left(\log^2 x + x \cdot (\log^2 x)' \right) = x^{x \log x} \cdot \left(\log^2 x + x \cdot \frac{2 \cdot \log x}{x} \right) \\ &= x^{x \log x} \cdot \log x \cdot (\log x + 2) \end{aligned}$$



Exercise 24

Compute the derivative of the following function:

$$f(x) = \frac{ax + b}{cx + d}$$

Using the rules for the derivative of a quotient, we have

$$f'(x) = \frac{a(cx+d) - c(ax+b)}{(cx+d)^2} = \frac{acx + ad - acx - bc}{(cx+d)^2} = \frac{ad - bc}{(cx+d)^2}$$

Exercise 25

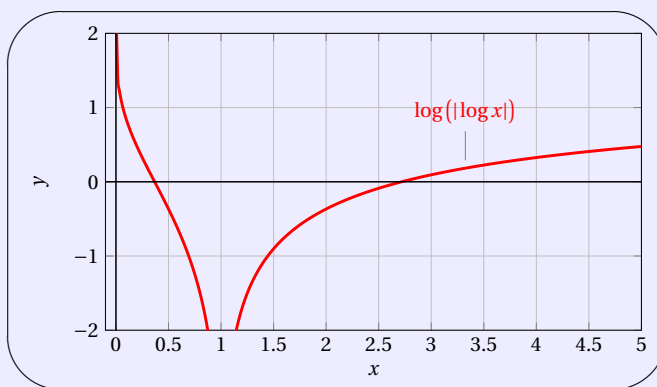
Compute the derivative of the following function:

$$f(x) = \log(|\log x|)$$

For $x \neq 1$, we have:

$$f(x) = \begin{cases} \log(\log x), & x > 1 \\ \log(-\log x), & x < 1 \end{cases} \quad \text{hence} \quad f'(x) = \begin{cases} \frac{1}{\log x} \cdot \frac{1}{x} = \frac{1}{x \log x}, & x > 1 \\ \frac{1}{-\log x} \cdot -\frac{1}{x} = \frac{1}{x \log x}, & x < 1 \end{cases}$$

The function is discontinuous at $x = 1$, hence it is not differentiable.



Exercise 26

Compute the derivative of the following function f , specifying the domain of f and of f' :

$$f(x) = \sqrt[4]{x^2 \log(x^3)}$$

From $x^3 > 0$ and $x^2 \log(x^3) \geq 0$ it follows that f is defined for $x \in [1, +\infty)$, and for such x we have

$$f(x) = \sqrt[4]{3x^2 \log x}$$

By the usual differentiation rules, for $x > 1$ the function f is differentiable and we have:

$$f'(x) = \frac{\sqrt[4]{3}}{4\sqrt[4]{x^6 \log^3 x}} (2x \log x + x).$$

For $x = 1$, since the usual differentiation rules cannot be applied (the function $\sqrt[4]{y}$ is not differentiable at $y = 0$), we directly examine the limit of the difference quotient, where the increment h makes sense only if $h > 0$:

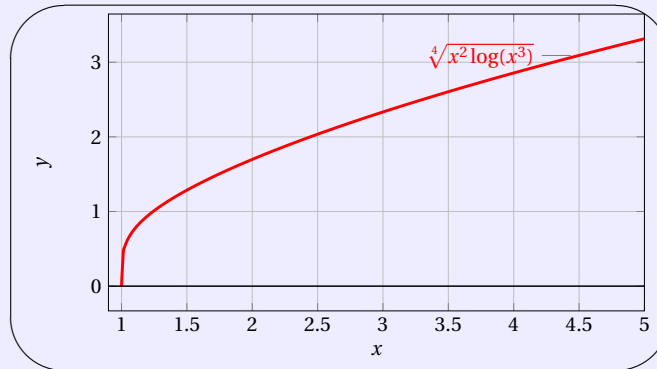
$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} &= \sqrt[4]{3} \lim_{h \rightarrow 0^+} \frac{\sqrt[4]{(1+h)^2 \log(1+h)}}{h} = \\ &= \sqrt[4]{3} \lim_{h \rightarrow 0^+} \frac{\log^{1/4}(1+h)}{h} = \sqrt[4]{3} \lim_{h \rightarrow 0^+} \frac{h^{1/4}}{h} = \sqrt[4]{3} \lim_{h \rightarrow 0^+} \frac{1}{h^{3/4}} = +\infty \end{aligned}$$

where we used $\log^{1/4}(1+h) \sim h^{1/4}$ for $h \rightarrow 0^+$. The function f is not differentiable at $x = 1$. At the point $(1, 0)$ the graph has a vertical tangent line.

Alternatively, we reach the same conclusion $f'(1) = +\infty$ by observing that f is continuous and that

$$\lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1^+} f'(x) = \lim_{x \rightarrow 1^+} \frac{\sqrt[4]{3}}{4\sqrt[4]{x^6 \log^3 x}} (2x \log x + x) = +\infty$$

using Lagrange's mean value theorem or, equivalently, L'Hôpital's rule for the indeterminate forms $0/0$.



Exercise 27

Give an example of $f : \mathbb{R} \rightarrow \mathbb{R}$, continuous at $x = 1$, not differentiable at $x = 1$, and such that $f(0) = 2$.

For example

$$f(x) = |x - 1| + 1.$$

The function satisfies $f(0) = 2$ and is continuous on all of $\mathbb{R}$.

At $x = 1$ we have $f'_+(1) = 1$, $f'_-(1) = -1$, hence the function is not differentiable.

